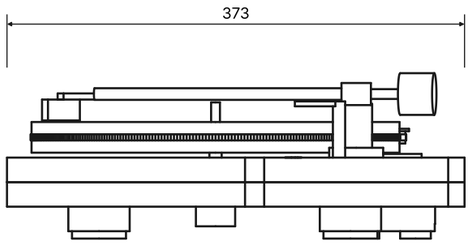
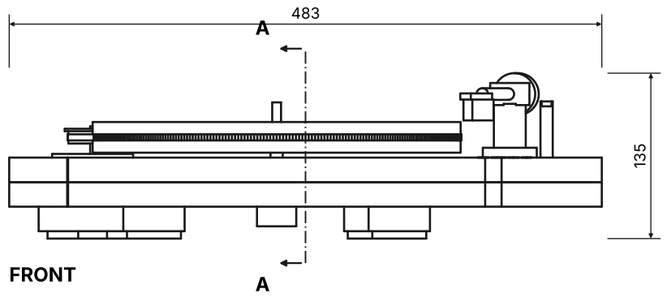
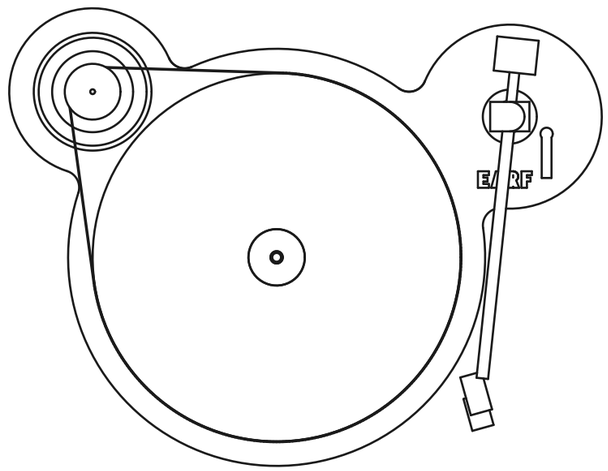
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

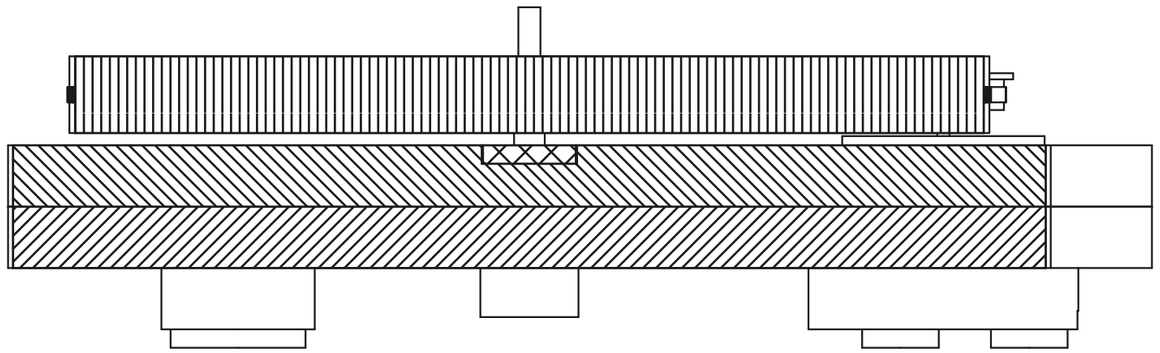
General arrangement

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. Section A-A (sheet 1): the product cut at x 241.5 and seen from the right; cut faces hatched, a different angle for each part.



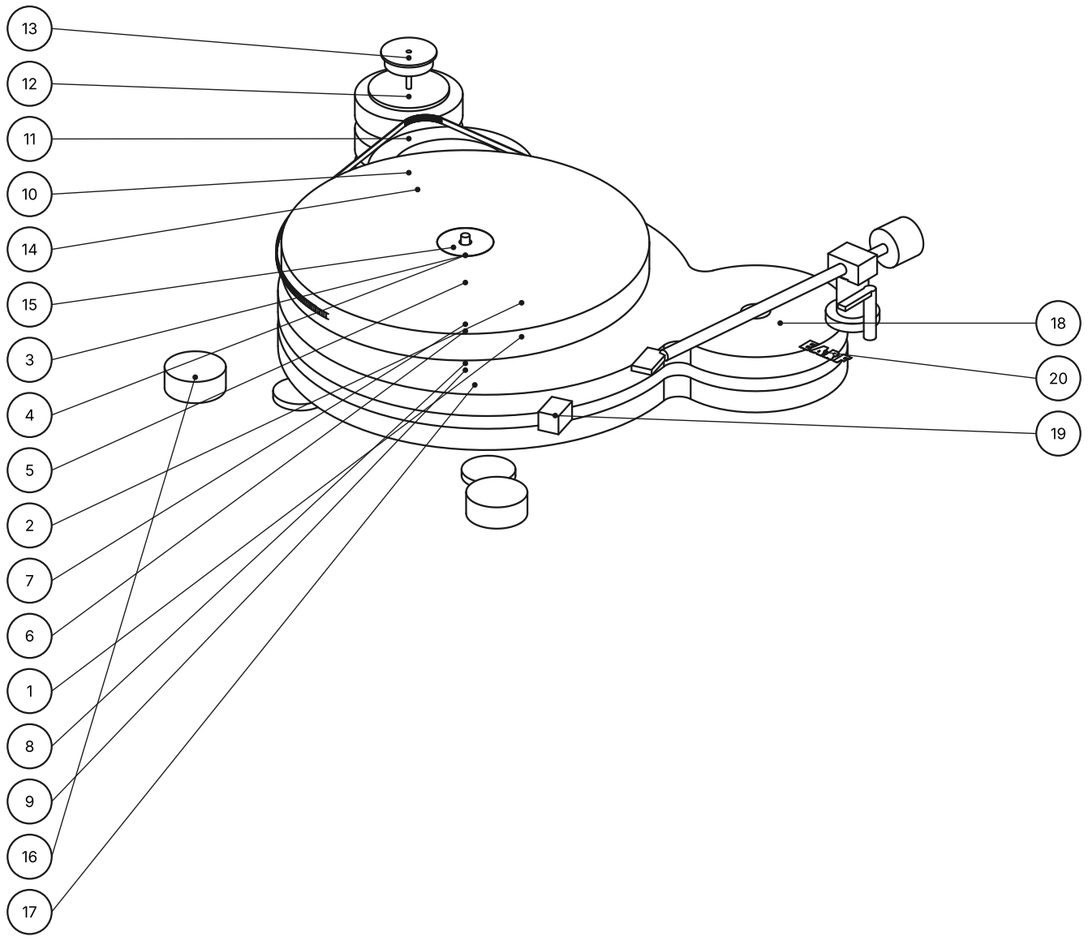
Section A-A

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5, parts drawn apart. Balloons are the parts list's numbers (last sheet).



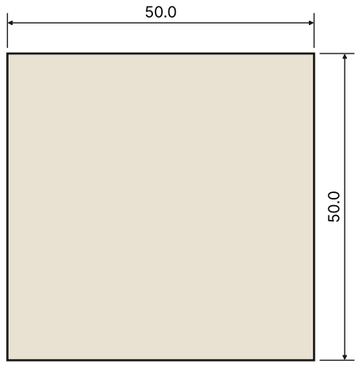
Exploded view

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



foot-1, foot-2, foot-3

20 mm cast-acrylic, 3 for the set

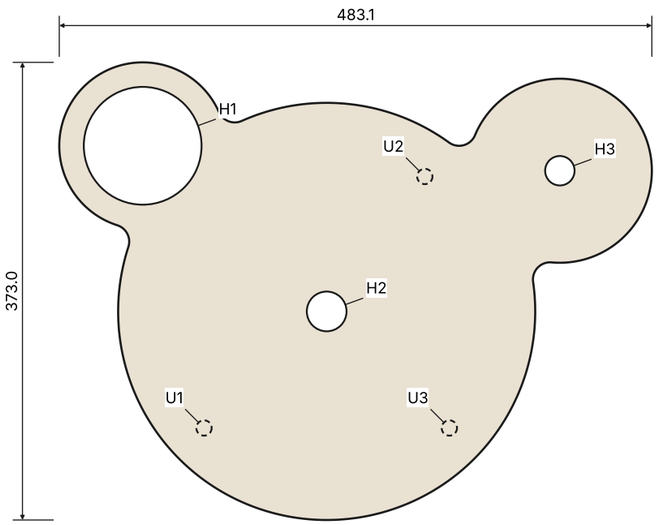
Sheet parts: plain panels

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/plinth-lower.dxf. Marks listed at the right.



plinth-lower: 20 mm cast-acrylic, 1 for the set, seen from its outer face

inner-face pockets: dxf/plinth-lower-underside.dxf (seen from that face)

mark	size	x	y	depth
H1	Ø96.0	68.0	305.0	through
H2	Ø32.4	218.0	170.0	through
H3	Ø24.0	408.1	284.7	through
U1	Ø12.5	118.0	75.0	15 from inside
U2	Ø12.5	298.0	280.0	15 from inside
U3	Ø12.5	318.0	75.0	15 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

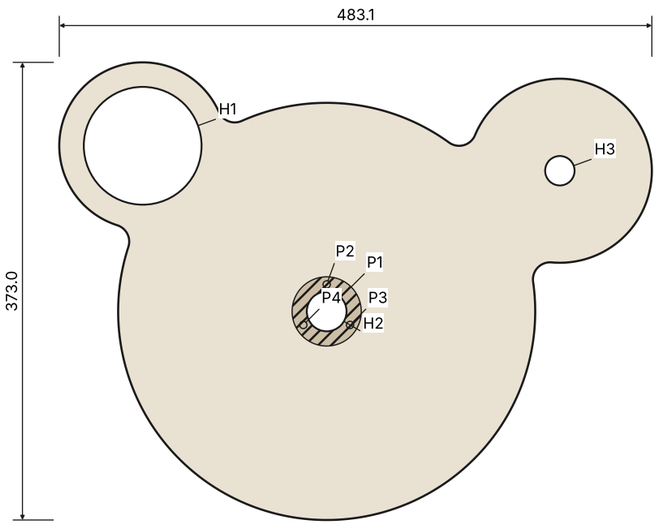
Sheet part: plinth-lower

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/plinth-upper.dxf. Marks listed at the right.



plinth-upper: 20 mm cast-acrylic, 1 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø56.4	218.0	170.0	6 from face
P2	Ø6.0	218.0	192.0	16 from face
P3	Ø6.0	237.1	159.0	16 from face
P4	Ø6.0	198.9	159.0	16 from face
H1	Ø96.0	68.0	305.0	through
H2	Ø32.4	218.0	170.0	through
H3	Ø24.0	408.1	284.7	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

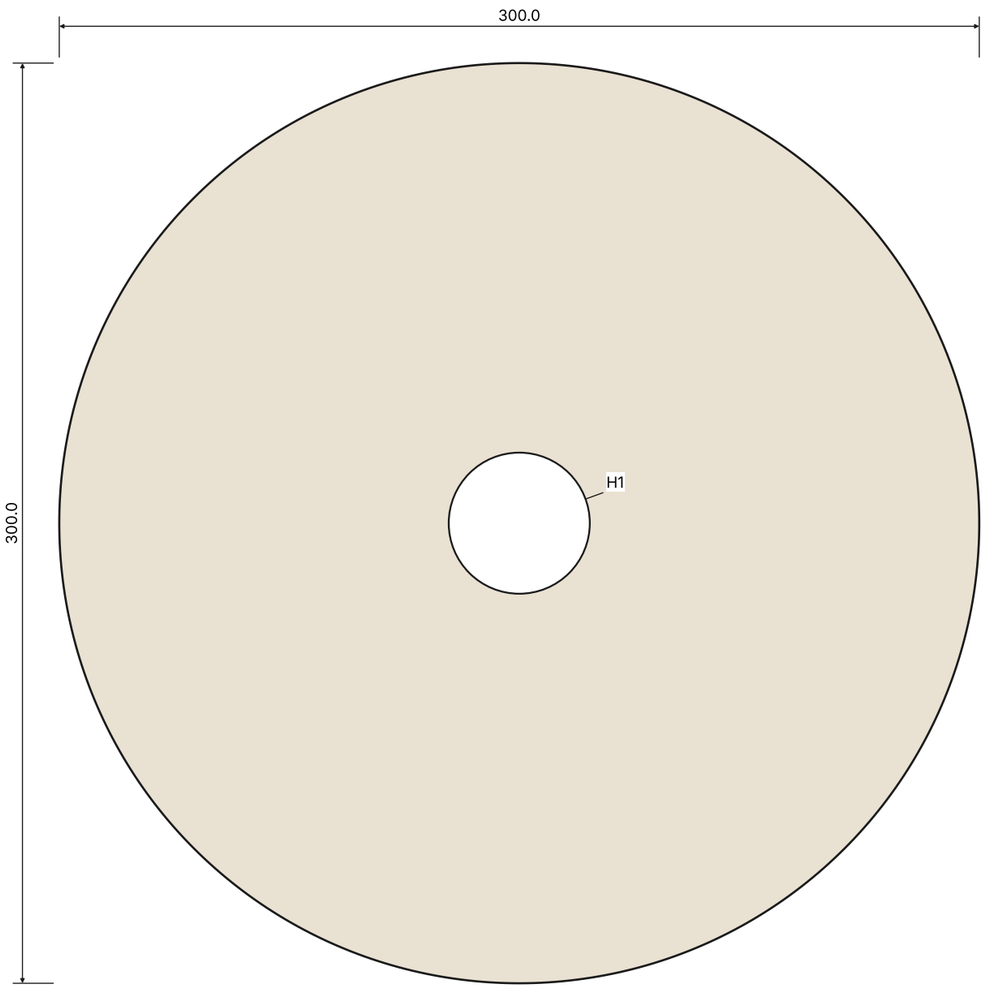
Sheet part: plinth-upper

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/platter.dxf. Marks listed at the right.



platter: 25 mm cast-acrylic, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø46.0	150.0	150.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

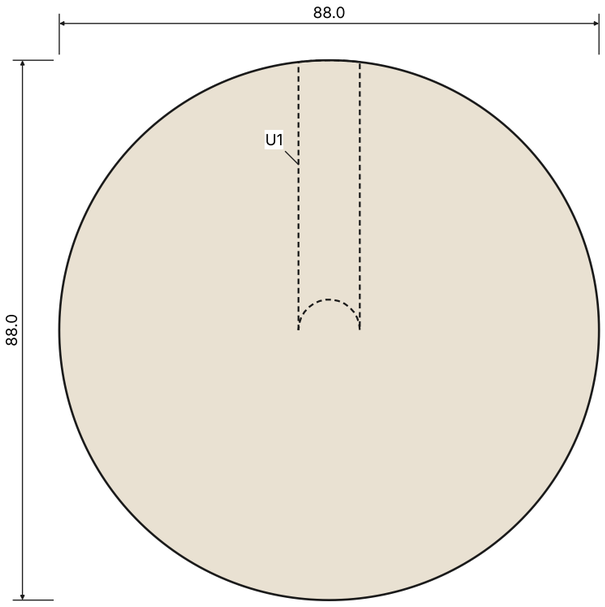
Sheet part: platter

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/pod-1.dxf. Marks listed at the right.



pod-1: 20 mm cast-acrylic, 1 for the set, seen from its outer face

inner-face pockets: dxf/pod-1-underside.dxf (seen from that face)

mark	size	x	y	depth
U1	profile (DXF)	44.0	66.0	6 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: pod-1

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

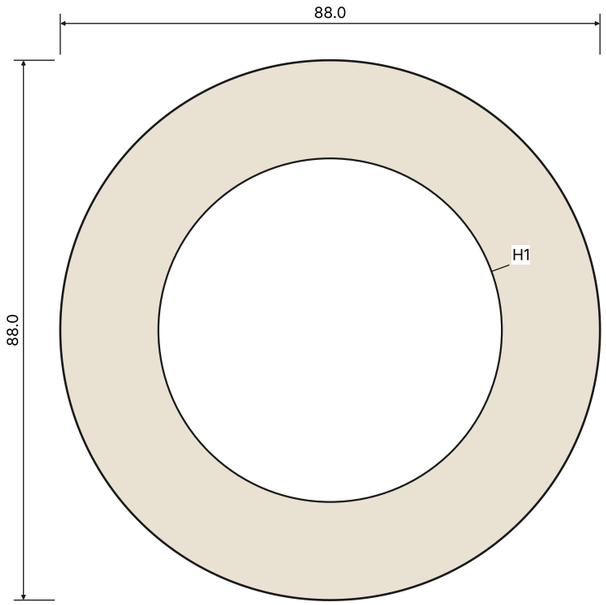
mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/pod-2.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø56.0	44.0	44.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



pod-2: 20 mm cast-acrylic, 1 for the set, seen from its outer face

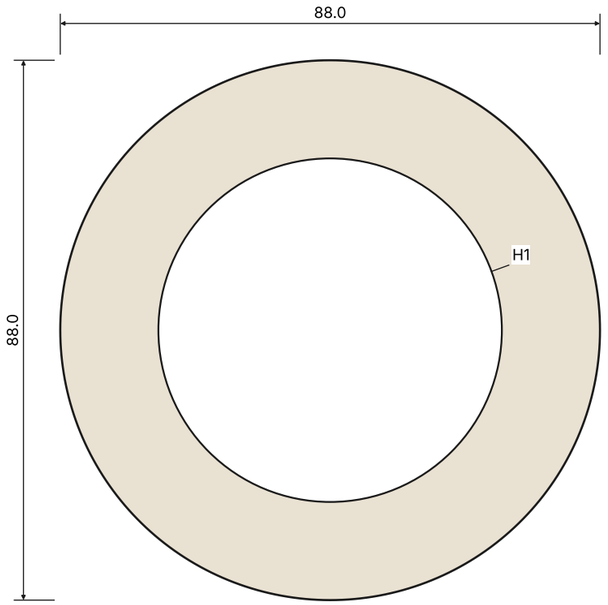
Sheet part: pod-2

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/pod-3.dxf. Marks listed at the right.



pod-3: 20 mm cast-acrylic, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø56.0	44.0	44.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: pod-3

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

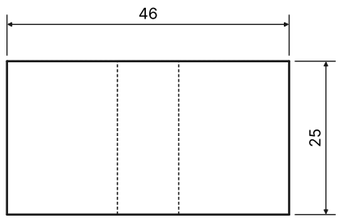
mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

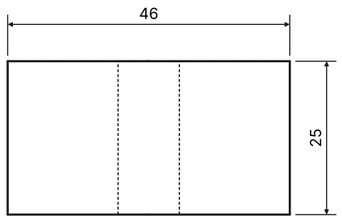
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

4. platter-hub: lathe, brass; 40 cm3 each; 1 for the set; 1:1

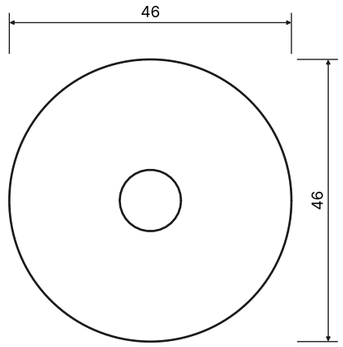
bonded into the platter with a structural acrylic adhesive; bored after, true to the spindle



FRONT



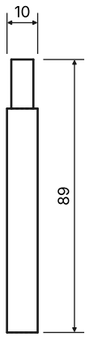
FROM THE RIGHT



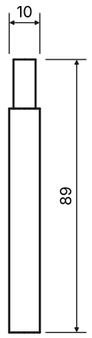
FROM ABOVE

5. spindle: lathe, stainless; 6 cm3 each; 1 for the set; 1:2

440C or hardened 416, ground to 10 h5; the record spindle 7.2; a 6 cup at its foot for the ball; pressed and bonded into the hub



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

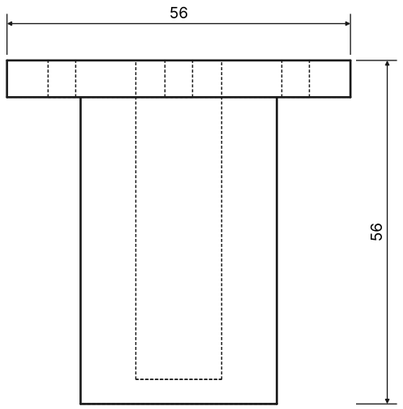
mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

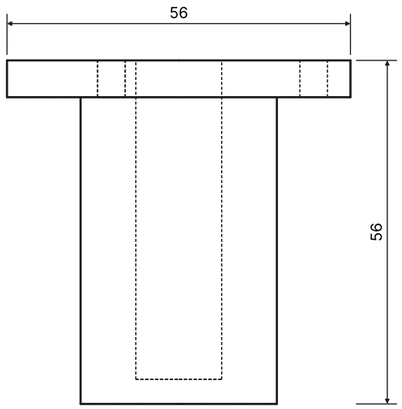
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

6. bearing-housing: lathe, brass; 47 cm3 each; 1 for the set; 1:1

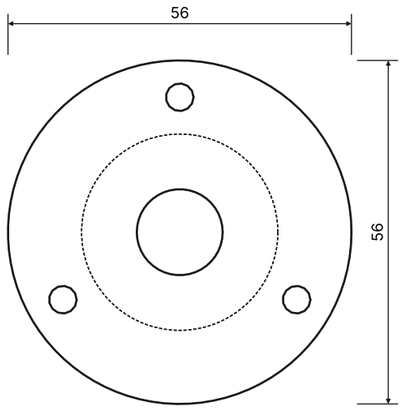
C360 brass; the bushes pressed in and reamed in line; filled with 2 ml of oil



FRONT



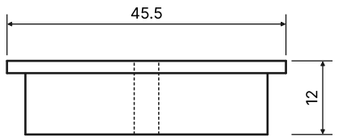
FROM THE RIGHT



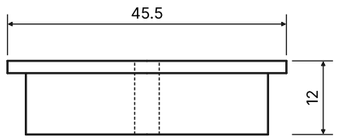
FROM ABOVE

13. pulley: lathe, aluminium; 15 cm3 each; 1 for the set; 1:1

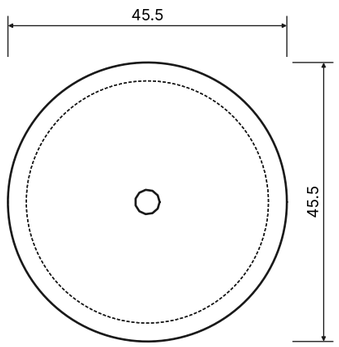
its belt face 39.48 at 250 rpm (Neo, 33.3); turned to the motor's measured speed; the Neo trims the rest



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	plinth-lower	cut	1	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	483 × 373	85.59
2	plinth-upper	cut	1	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	483 × 373	85.59
3	platter	cut	1	25 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	300 × 300	53.44
4	platter-hub	lathe	1	Brass C360	56 × 56 × 35	58.17
5	spindle	lathe	1	Stainless steel 304	20 × 20 × 99	126.19
6	bearing-housing	lathe	1	Brass C360	66 × 66 × 66	79.50
7	bushes	buy	1	Oil-impregnated sintered bronze plain bushes 10 × 14 × 12, a pair (Oilite kind), reamed to a 0.01 running fit on the spindle after pressing		8.00
8	ball	buy	1	Tungsten carbide ball, 6 mm, grade 10: the spindle's thrust point, seated in a cup at the spindle's foot		5.00
9	thrust-pad	buy	1	Thrust pad, PTFE (or PEEK), 13.8 × 2, at the bottom of the housing's bore, under the ball		3.00
10	pod-1	cut	1	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	88 × 88	3.68
11	pod-2, pod-3	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	88 × 88	3.68
12	motor	buy	1	Rega 24 V low-vibration synchronous motor upgrade kit (motor, its board and mounting), for the Neo PSU		200.00
13	pulley	lathe	1	Aluminium 6061	55 × 55 × 22	46.69
14	pod-pads	buy	1	Isolate It (or alike) Sorbothane discs 25 × 6, Shore 30, a set of 3, under the motor's pod (about 0.6 kg on them: soft, so the motor's hum stays in the pod)		15.00
15	belt	buy	1	Flat turntable belt, ground rubber or polyurethane, 5 wide x 0.6, about 975 long unstretched (1025 round the platter's rim and the 40 pulley, stretched 5 %)		25.00
16	foot-1, foot-2, foot-3	cut	3	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	50 × 50	1.19
17	foot-pads	buy	1	Isolate It (or alike) Sorbothane hemispheres or discs, 44 across, 6 tall, Shore 50 or 70 (chosen by the loaded weight, about 2 kg a foot), a set of 3, bonded under the plinth's feet		30.00
18	arm	buy	1	Rega RB330 tonearm (9 in, Rega geometry: 222 mm from the platter's centre to the arm's hole, effective length 236.5, overhang 14.5, offset 21.55 deg), with its own lead and cueing lift		675.00
19	cartridge	buy	1	Ortofon 2M Blue moving-magnet cartridge (nude elliptical stylus, 5.5 mV), 1/2 in mount		220.00
20	mark	buy	1	Laser etching into the cast acrylic, frosted, about 0.1 deep, from a vector file: EARF 44 wide, the etch filled with the colour, deeper		15.00
21	neo-psu	buy	1	Rega Neo PSU MK2: DSP-generated drive for Rega's 24 V motor, 33/45 by a button, fine speed trim (0.01 rpm steps) on the back; US 60 Hz 115-120 V version		445.00
22	feet-studs	buy	1	Three M8 brass threaded inserts bonded with epoxy into the plinth's bottom sheet (12.5 holes, 15 deep), three M8 × 30 brass studs and lock nuts: the feet turn on them to level the platter		14.00
23	bearing-fixings	buy	1	Three M4 brass threaded inserts bonded with epoxy into 6 holes in the pocket's floor, three M4 × 10 black socket button heads: the bearing's flange		5.00
24	bearing-oil	buy	1	Turntable bearing oil, 8 ml (a light synthetic oil, ISO VG 32 or as the bush maker says)		15.00

Parts list (1 of 2)

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
25	cement	buy	1	Two-part polymerising acrylic cement for cast sheet (Acrifix 1R 0192 kind), clear: the cemented layers (a 100 g kit does one piece)		20.00

Parts list (2 of 2)

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. The plinth: two 20 mm sheets of tinted cast acrylic (the Clouds' own, in the colour), CNC cut as one outline of three circles, its corners R14 and its edges eased R2 and polished, cemented with a two-part polymerising acrylic cement (Acrifix 1R 0192 kind) on a flat table, annealed after (about 80 C, an hour for each 3 mm, cooled slowly).
2. The motor stands apart: its pod (three 88 discs cemented) on three soft Sorbothane discs on the shelf, 4 mm clear of the plinth all round. Nothing touches the plinth from the motor but the belt.
3. The platter: 25 mm of the same acrylic, 300 across, its rim polished and crowned; the brass hub bonded in, the spindle pressed and bonded into the hub, the whole balanced. No mat.
4. The bearing: a turned brass housing through both sheets, flanged into a pocket in the top on three M4 screws into bonded inserts; two sintered bronze bushes; a ground stainless spindle on a 6 mm carbide ball on a PTFE pad. Oiled.
5. Speed: Rega's 24 V motor driven by the Neo PSU MK2 (33/45 by a button, 0.01 rpm trim); the pulley turned to the motor's measured speed so the Neo trims from the middle of its range; a flat belt on the platter's rim.
6. Arm: Rega RB330, 222 from the spindle (Rega geometry), its nut under the plinth through a 24 hole; an Ortofon 2M Blue. Level the platter on the three feet (each turns on its M8 stud) before setting the arm.
7. The mark: EARF 44 wide laser-etched in the arm's circle in front of the arm and filled with the colour, deeper.

Notes

Cumulus: Belt-drive record player in tinted cast acrylic, the motor apart

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15